

LOAN APPROVAL PREDICTION USING MACHINE LEARNING

Executive Summary

The objective of this project was to develop a machine learning model capable of predicting loan approval decisions based on applicant information. Financial institutions process thousands of loan applications and must assess applicant eligibility accurately while minimizing risk. Manual evaluation can be time-consuming and inconsistent.

This project utilized a Loan Approval Prediction dataset containing demographic, financial, and credit-related information of applicants. Data preprocessing techniques such as missing value handling, categorical encoding, feature scaling, and class imbalance handling using SMOTE were applied. Multiple machine learning models were trained and evaluated, including Logistic Regression, Decision Tree, and Random Forest.

The results showed that Logistic Regression achieved the best overall performance with a ROC-AUC score of 0.833 and an F1 Score of 0.888. Feature importance analysis revealed that Credit History, Applicant Income, and Loan Amount are the most influential factors affecting loan approval decisions.

The developed solution demonstrates how machine learning can support financial institutions in automating loan approval processes, reducing risk, and improving decision-making efficiency.

1. Project Objective

The objective of this project is to build a supervised machine learning model capable of predicting whether a loan application should be approved or rejected based on applicant characteristics.

The project aims to:

- Analyze borrower information.
- Handle missing values and data quality issues.
- Encode categorical variables.
- Address class imbalance using SMOTE.
- Train and compare multiple machine learning models.
- Evaluate models using Precision, Recall, F1 Score, and ROC-AUC.
- Provide business-oriented insights and recommendations.

2. Dataset Description

The dataset contains loan applicant information used by financial institutions during the loan approval process.

Dataset Information

- Total Records: 614
- Total Features: 13
- Target Variable: Loan_Status

Features Included

- Gender
- Married
- Dependents
- Education
- Self Employed
- Applicant Income
- Coapplicant Income
- Loan Amount
- Loan Amount Term
- Credit History
- Property Area
- Loan Status

Target Variable

- Y = Loan Approved
 - N = Loan Rejected
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3. Data Preprocessing

Data preprocessing was performed to prepare the dataset for machine learning.

3.1 Missing Value Handling

Several columns contained missing values:

- Credit_History
- Self_Employed
- LoanAmount
- Dependents
- Loan_Amount_Term
- Gender

- Married

Approach

- Numerical columns were filled using Median values.
- Categorical columns were filled using Mode values.

This ensured that no records were lost while maintaining data integrity.

3.2 Feature Encoding

Machine learning algorithms require numerical input.

Categorical variables were converted into numerical representations using Label Encoding.

Examples:

- Male $\rightarrow$ 1
 - Female $\rightarrow$ 0
 - Yes $\rightarrow$ 1
 - No $\rightarrow$ 0
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3.3 Feature Scaling

StandardScaler was applied to numerical variables to normalize feature distributions.

Benefits:

- Faster model convergence.
 - Improved model performance.
 - Reduced feature scale bias.
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3.4 Handling Class Imbalance

The target variable exhibited class imbalance.

Before SMOTE

- Approved Loans: 337
- Rejected Loans: 154

After SMOTE

- Approved Loans: 337
- Rejected Loans: 337

SMOTE was used to synthetically generate minority-class samples and create a balanced training dataset.

4. Model Development

Three machine learning models were trained and evaluated.

4.1 Logistic Regression

Logistic Regression is a linear classification algorithm commonly used in credit risk and loan approval applications.

Results

- Accuracy: 83.74%
 - Precision: 84.95%
 - Recall: 92.94%
 - F1 Score: 88.76%
 - ROC-AUC: 83.34%
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4.2 Decision Tree

Decision Trees provide interpretable rule-based predictions.

Results

- Accuracy: 70.73%
 - Precision: 83.56%
 - Recall: 71.76%
 - F1 Score: 77.21%
 - ROC-AUC: 70.09%
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4.3 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees.

Results

- Accuracy: 80.49%
- Precision: 84.27%
- Recall: 88.24%
- F1 Score: 86.21%

- ROC-AUC: 77.24%

5. Model Comparison

Model	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.849	0.929	0.888	0.833
Random Forest	0.843	0.882	0.862	0.772
Decision Tree	0.836	0.718	0.772	0.701

Best Model

Based on the evaluation metrics, Logistic Regression emerged as the best-performing model.

Reasons:

- Highest Recall
- Highest F1 Score
- Highest ROC-AUC
- Strong balance between precision and recall

6. Feature Importance Analysis

Feature importance analysis identified the most influential factors affecting loan approval decisions.

Top Factors

1. Credit History
2. Applicant Income
3. Loan Amount
4. Coapplicant Income
5. Property Area

Observation

Credit History was the strongest predictor of loan approval, indicating that previous repayment behavior significantly influences lending decisions.

7. Business Insights

The analysis generated several important business insights.

Insight 1

Applicants with positive credit histories have a significantly higher probability of loan approval.

Insight 2

Higher applicant income increases approval likelihood.

Insight 3

Large loan amounts may reduce approval probability depending on applicant financial strength.

Insight 4

Property area contributes to approval decisions, indicating geographical influence on lending risk.

Insight 5

Balanced datasets improve predictive performance and fairness in loan decision systems.

8. Deployment Recommendation

The default classification threshold is 0.50.

For deployment purposes, a threshold between 0.55 and 0.60 is recommended to balance business growth and risk management.

Threshold 0.50

- Higher approval rate
- Higher recall
- More customers approved

Threshold 0.60

- Lower risk
- More conservative approvals
- Improved confidence in decisions

Recommended Threshold: 0.55

9. Conclusion

This project successfully developed and evaluated machine learning models for predicting loan approval outcomes. Data preprocessing techniques, feature scaling, categorical encoding, and SMOTE-based balancing improved model performance.

Among all evaluated models, Logistic Regression achieved the best performance with a ROC-AUC score of 83.34% and an F1 Score of 88.76%.

The results indicate that Credit History, Income, and Loan Amount are the most influential factors affecting loan approval decisions. The developed model can assist financial institutions in improving decision-making efficiency, reducing manual effort, and managing lending risk more effectively.

This project demonstrates the practical application of machine learning in the financial services domain and highlights the importance of data-driven decision-making in loan approval systems.